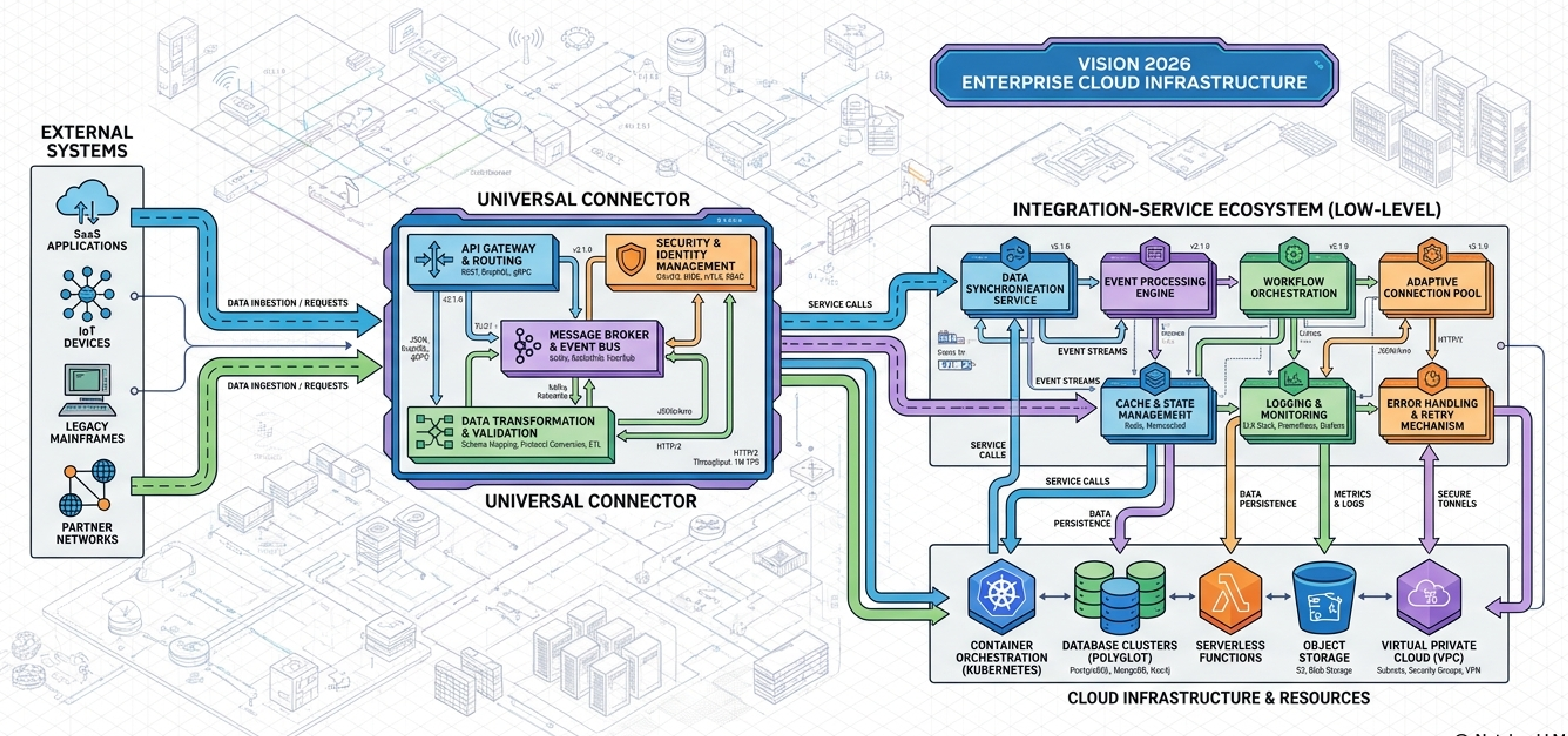
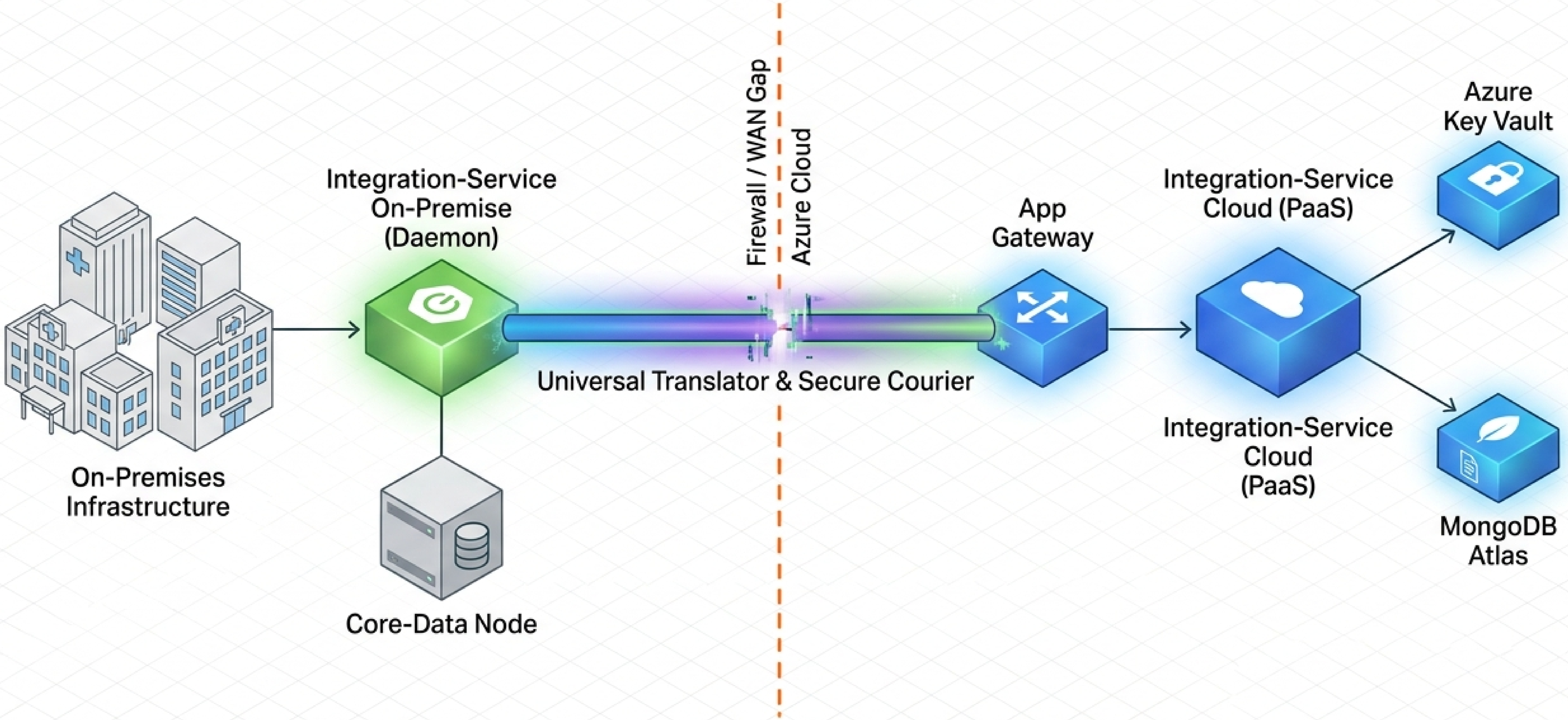
ARCHITECTING THE UNIVERSAL CONNECTOR

HIGH-LEVEL TO LOW-LEVEL DESIGN OF THE INTEGRATION-SERVICE ECOSYSTEM



The Bridge Across the Edge

The Integration-Service is a dedicated daemon process deployed on-premises at client facilities. It acts as an enterprise data universal connector, seamlessly and securely routing sanitized facility data to the Azure App-Core.



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Component Anatomy Map

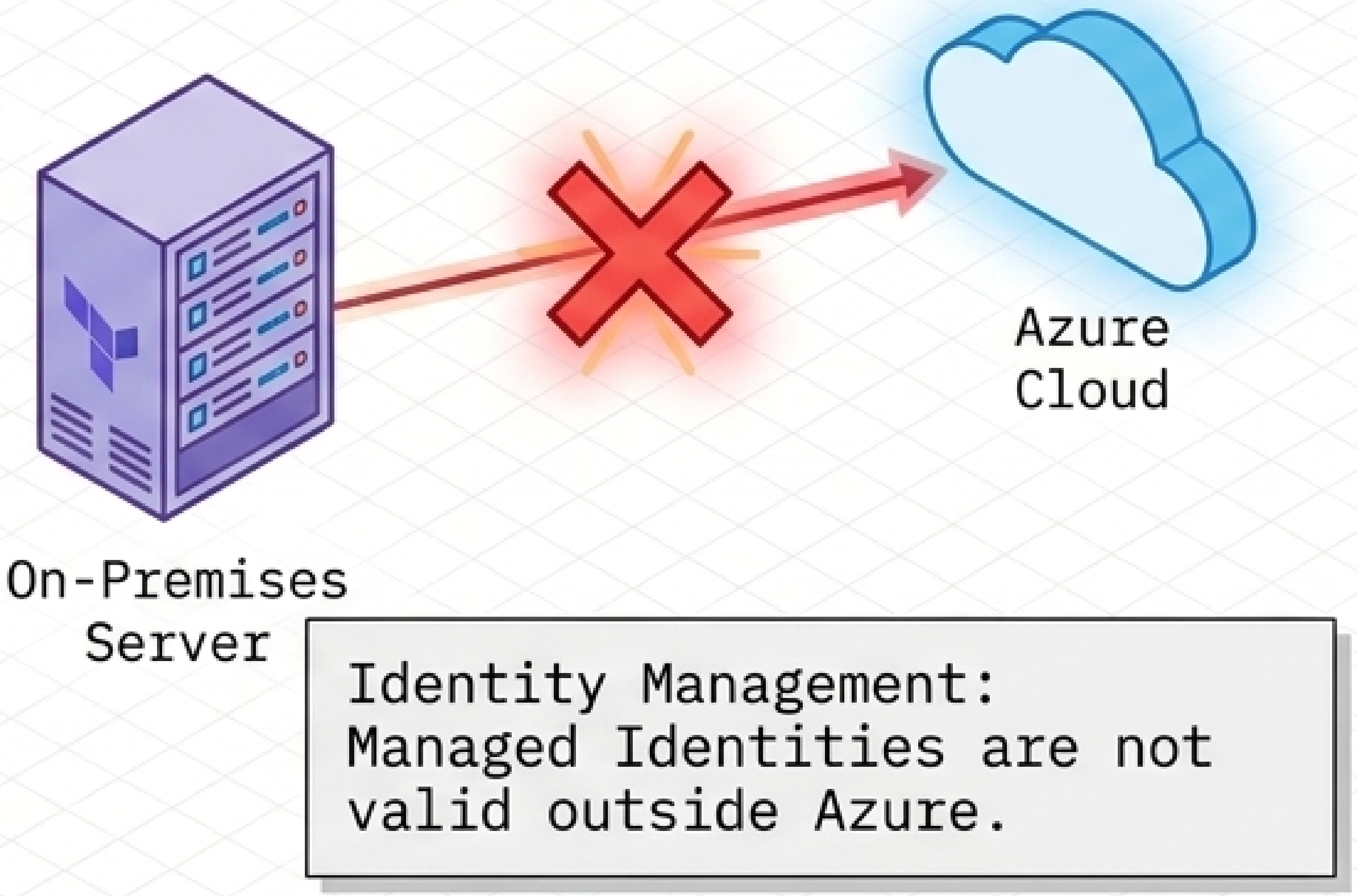
A decoupled architecture separating stateful on-premises data extraction from scalable, multi-tenant cloud persistence and identity governance.

	Location: On-Premise	Location: Azure Cloud
Actor: Enterprise Managed	Integration-Service On-Premise Stateful Daemon, App-Core Connector	Integration-Service-Cloud & App-Core Backend Multi-tenant APIs, Data Routers
Actor: Third-Party / Microsoft	Storage-Node Edge Image Storage	App Gateway Azure Key Vault Azure Files (SMB) MongoDB Atlas Scalable persistence and traffic orchestration

The Zero-Trust Identity Mandate

Because the daemon runs on-premises, it cannot leverage Azure Managed Identities. Instead, each deployed instance receives a unique, dynamically generated Azure AD App Registration, functioning as an independent Service Principal.

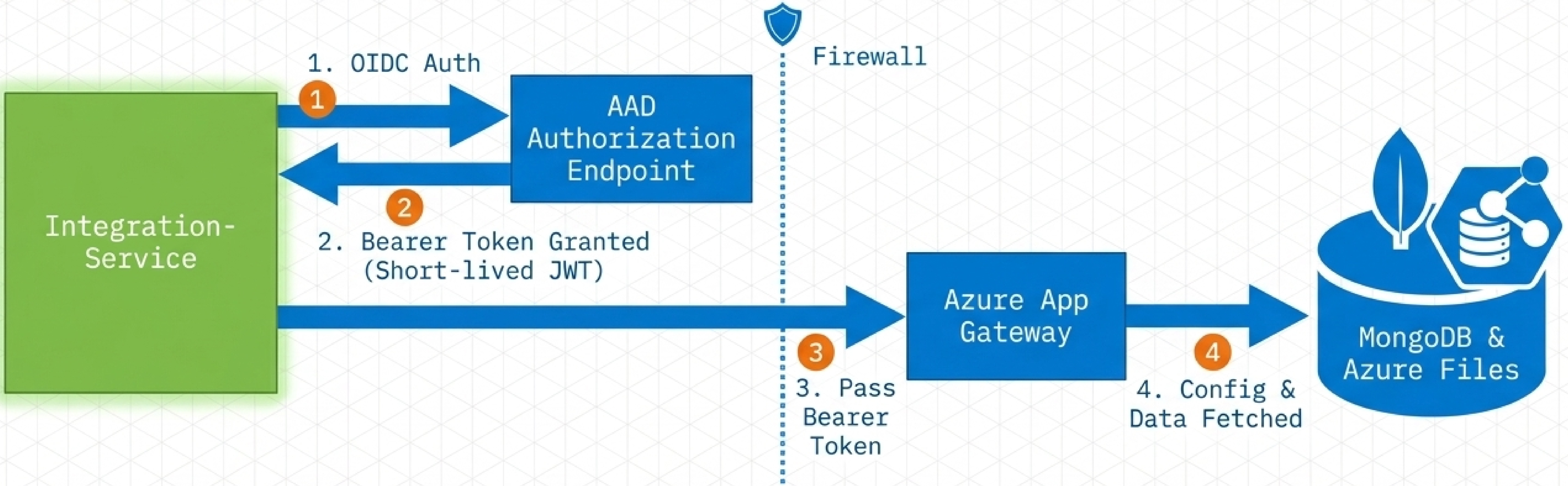
The Problem



The Solution

The Sequential Trust Handshake

The Integration-Service acts as an autonomous client. It authenticates via OpenID Connect (OIDC), exchanging its unique credentials for a short-lived Bearer Token to access the cloud data plane.



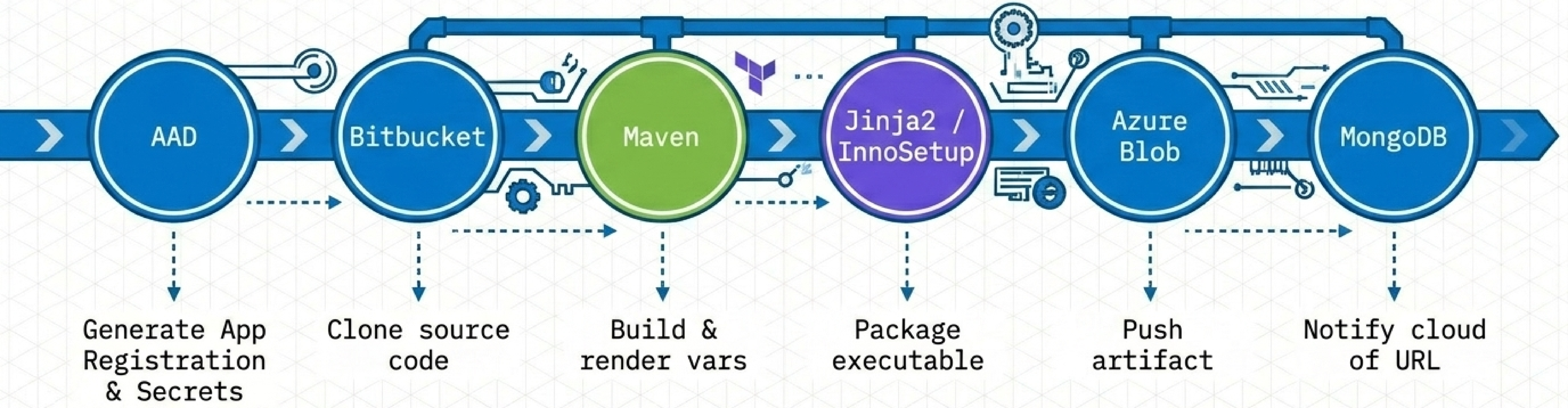
Strict Tier Parity (ENG vs. PRO)

Infrastructure development follows a strict parity model. The exact same deployment pipeline is used across tiers, with isolation enforced via physical subscription separation and branch-aware DNS routing.

Dimension	Engineering (ENG)	Production (PRO)
Azure Subscriptions	Enterprise DevTest Subscription	Enterprise Production Subscription
Pipeline Git Branch	develop	main
Example Endpoints	*.deng.enterprise.com	*.apps.enterprise.com
DevOps Service Connections	svccon-integration-dev (Contributor RBAC)	svccon-integration-pro (Contributor RBAC)

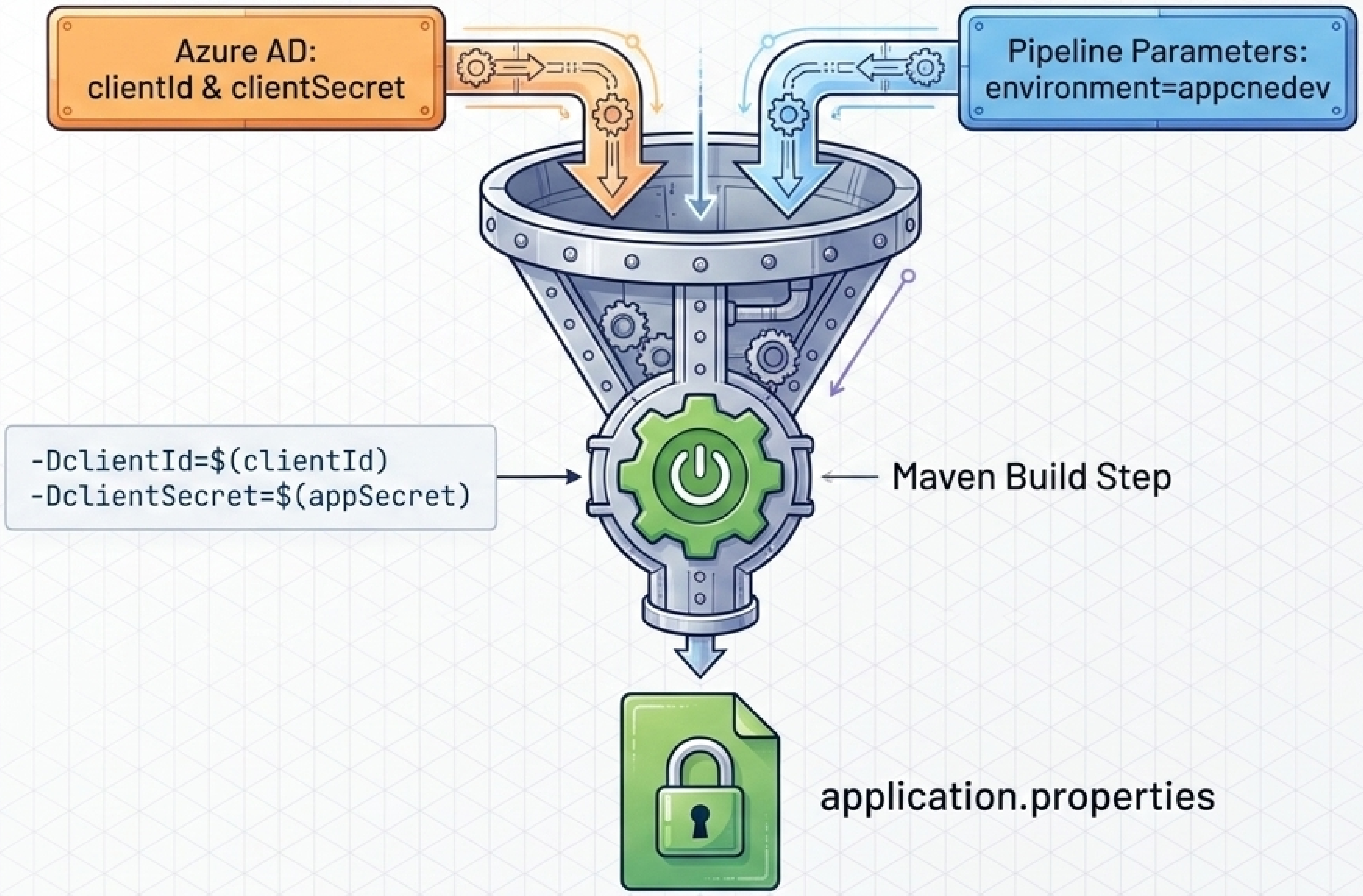
The CI/CD Assembly Line

Orchestrated via Azure DevOps pipelines, the build process manufactures a highly customized, secure Windows installer specific to a single client facility in real-time.



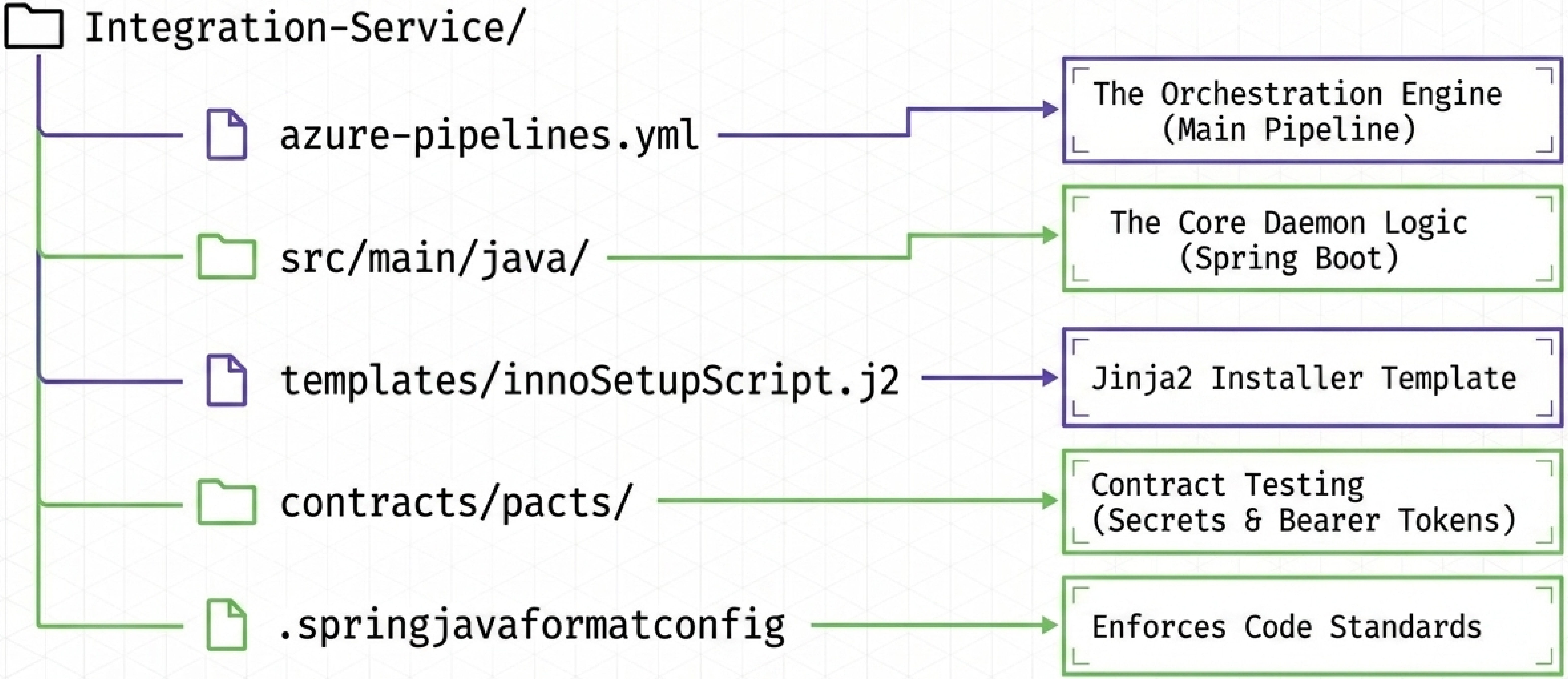
The Secret Injection Funnel

Credentials are treated as ephemeral assets. Dynamically generated App Registration secrets are injected directly into the Java properties during the Maven build phase, ensuring zero static credential exposure in the repository.



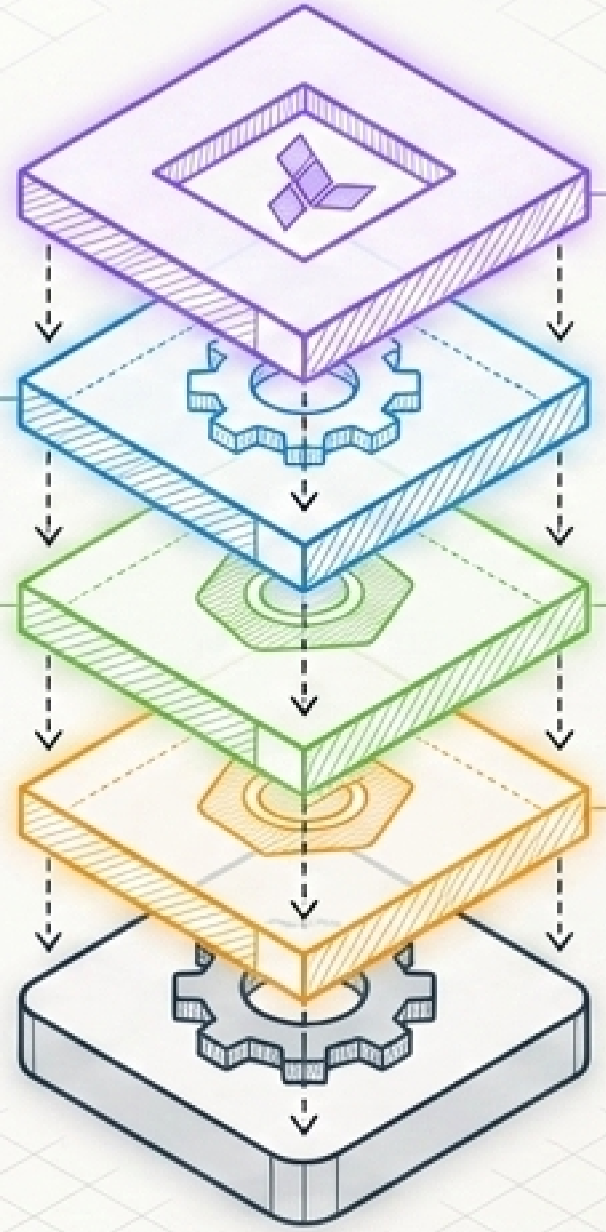
Code-as-Context: Repository Architecture

The repository encapsulates both the application logic and the infrastructure-as-code required to manufacture and package it.



Anatomy of a Custom Installer

The pipeline outputs a fully self-contained payload. It bundles the compiled Java artifact with a designated Java Runtime and a **Windows Service Wrapper**, ensuring a zero-dependency installation on the client's host machine.



The Wrapper `innoSetupScript.iss`
The external packaging configuration.

`winSW.exe` The Daemon
Windows Service Wrapper downloaded from Azure Artifacts.

The Runtime `JRE 11`
Java Runtime Environment from Azure Artifacts.

The Runtime `JRE 11`
Java Runtime Environment from Azure Artifacts.

The Core `Integration-Service.jar`
The compiled Spring Boot application logic.

`Integration-Service.exe`

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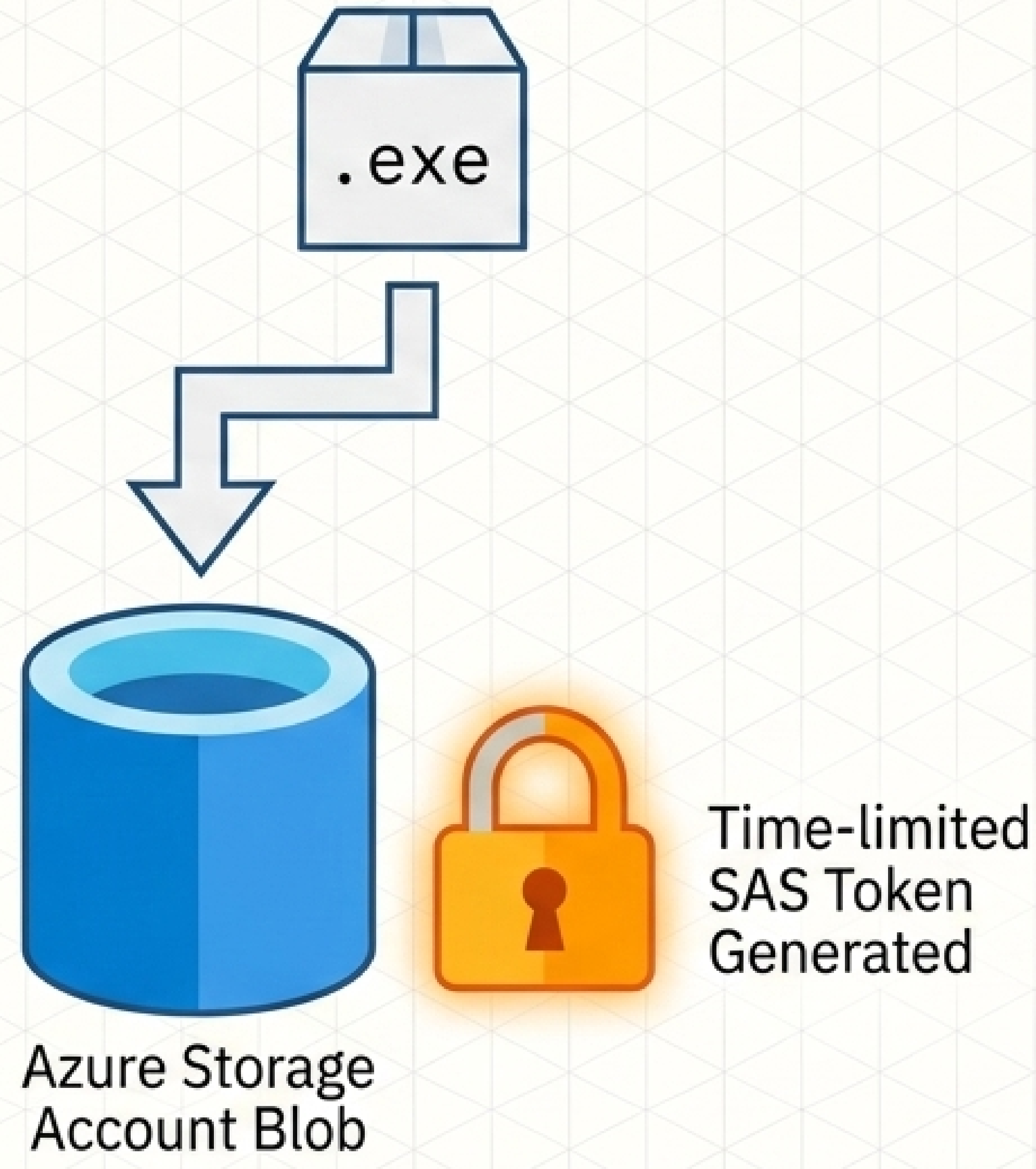
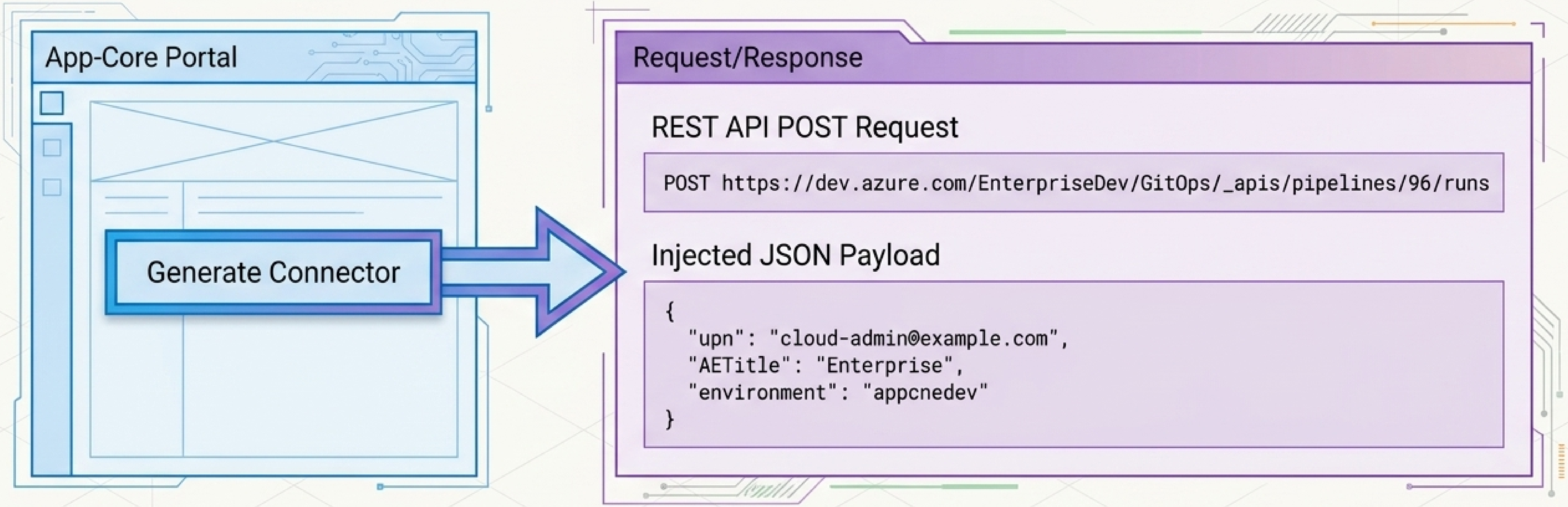
Artifact Delivery & Cloud Notification

Post-compilation, the installer is pushed to an Azure Blob container. The pipeline generates a secure, ephemeral SAS token and notifies the App-Core MongoDB, making the download immediately available via the Enterprise Portal.



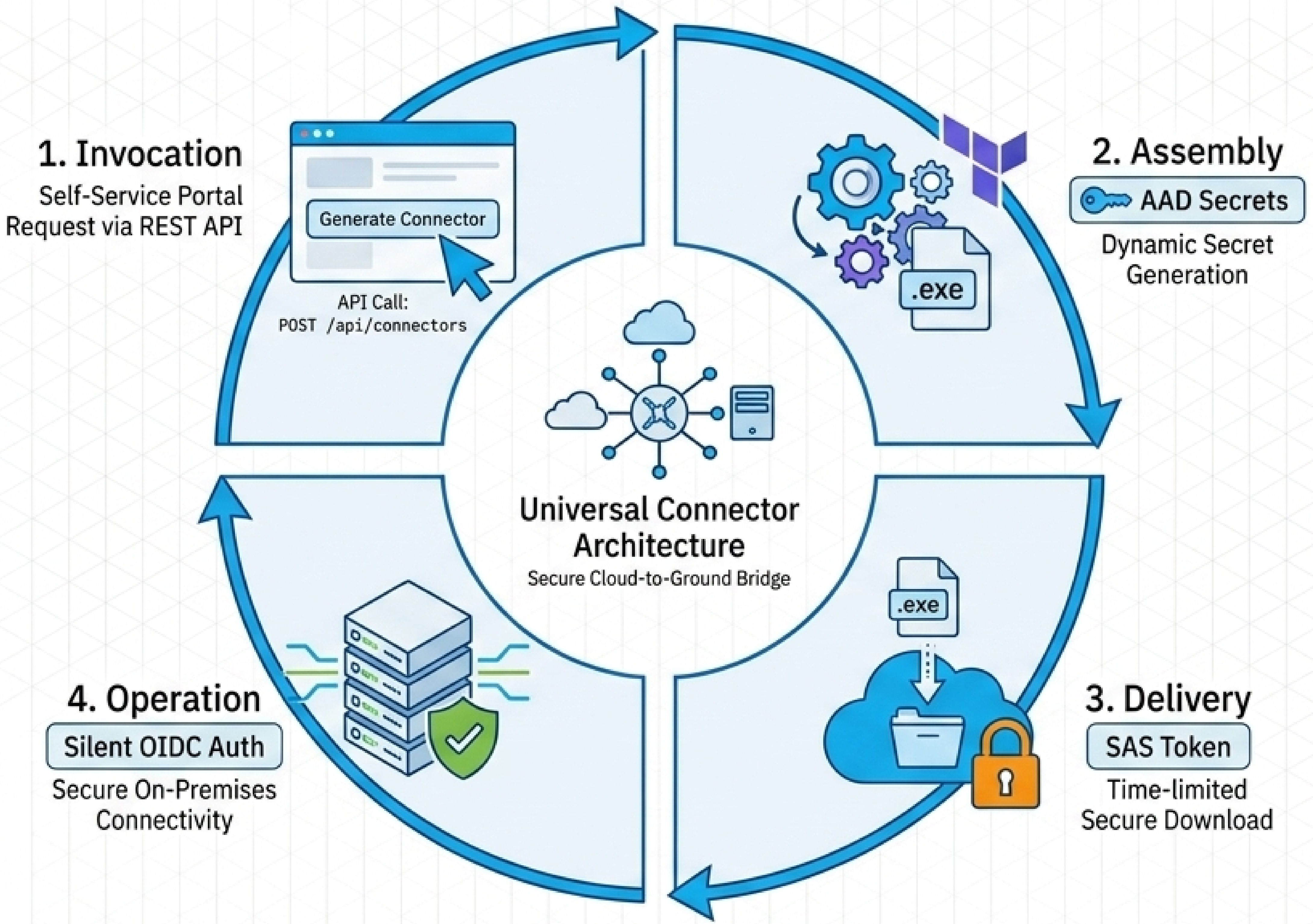
Invoking the Assembly Line

The entire DevSecOps pipeline operates as a **backend service**. It is triggered programmatically via a REST API call from the App-Core Portal, passing center-specific parameters to manufacture a custom installer on demand.



The Universal Connector Lifecycle

A fully automated, zero-trust ecosystem. From self-service API invocation to on-premises deployment, the Integration-Service pipeline eliminates static secrets and manual configuration, ensuring secure, multi-tenant enterprise connectivity at scale.



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